

Mobile Phone Price Category Prediction

Task 4 Completion Report: Data Cleaning and Preparation

Dataset: mobile_dataset.csv | Output: cleaned_dataset.csv

1. Task Objective

Clean the raw mobile phone data and prepare a reliable, model-ready dataset for Random Forest classification. The preparation preserves the price category target while excluding raw identifiers and leakage-prone fields from model inputs.

2. Cleaning Summary

Check	Result
Original records	1,019
Duplicate records removed	2
Records after cleaning	1,017
Model-ready columns	22
Missing values after preparation	0

3. Preparation Performed

Duplicate rows were removed. Numeric values were extracted from storage, battery, display, camera, and processor text. Connectivity indicators were converted into binary columns for 3G, 4G, 5G, VoLTE, Wi-Fi, and NFC. External-memory support, Android version, and FM-radio availability were also encoded.

Missing numeric values were imputed with the median of the cleaned data. Missing or unavailable binary values were represented as 0 where the source field did not indicate support. The target Price_Category was derived from price using the project price bands.

4. Model-Ready Dataset Design

Dataset component	Decision
Included	Numeric specifications, encoded connectivity features, and Price_Category
Excluded	Name and img because they are identifiers or URLs
Excluded	Raw specification text after useful values were extracted
Excluded	Spec_Score, Rating, and Is_Launched because they are summary/status fields, not required for prediction
Excluded	price because it directly creates Price_Category and would cause target leakage

5. Completion Status

Task 4 is complete. The cleaned and prepared dataset was generated successfully at cleaned_dataset.csv with 1,017 records, 22 columns, and no missing values. It is ready for train-test splitting and Random Forest model development.